

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc Microbiology/ Biotechnology/ Biochemistry Examination April/ May 2018

Course code & Name BS 208 & General Microbiology

Date: 7/5/2018 Day: Monday Time: 10:00 am to 10:30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Study of fungi is called as ____		01
	(i) Mycology	(ii) Phycology	
	(iii) Micology	(iv) None of the above	
2.	_____ disapproved concept of abiogenesis very first time.		01
	(i) John Needham	(ii) Louis Pasteur	
	(iii) Francesco Redi	(iv) Alexander Fleming	
3.	Who termed the name 'Vaccine'?		01
	(i) Edward Jenner	(ii) Loius Pasteur	
	(iii) Francesco Redi	(iv) Alexander Fleming	
4.	In Gram staining, ____ acts as decolorizer.		01
	(i) Crystal violet	(ii) Iodine	
	(iii) Alcohol	(iv) Safranin	

5.	_____ can be used to preserve microbial cultures for years.		01
	(i) Lyophilization	(ii) Agar slants	
	(iii) Liquid nitrogen	(iv) Both, (i) and (iii)	
6.	Nigrosin is _____		01
	(i) Acidic stain	(ii) Basic stain	
	(iii) Cationic stain	(iv) Both, (ii) and (iii)	
7.	Penicillin was discovered by _____		01
	(i) Edward Jenner	(ii) Louis Pasteur	
	(iii) Francesco Redi	(iv) Alexander Fleming	
8.	_____ uses carbon dioxide as sole source of carbon.		01
	(i) Lithotrophs	(ii) Heterotrophs	
	(iii) Autotrophs	(iv) Phototrophs	
9.	_____ can be used to study organism in the living state.		01
	(i) Dark field Microscopy	(ii) Phase contrast Microscopy	
	(iii) Scanning electron Microscopy	(iv) Both, (i) and (ii)	
10.	The time required to reduce the microbial population by 90% at a specified temperature and specified conditions is called as _____		01
	(i) D-value	(ii) F-value	
	(iii) Z-value	(iv) None of the above	
11.	_____ is sterilization by moist heat under pressure.		01
	(i) Hot Air Oven	(ii) Incineration	
	(iii) Filtration	(iv) Autoclave	
12.	Bacteria growing at high NaCl are called as _____		01
	(i) Thermophiles	(ii) Barophiles	
	(iii) Halophiles	(iv) Neutrophiles	
13.	Bacteria produce antibiotics during _____ phase.		01
	(i) Lag	(ii) Log	
	(iii) Stationary	(iv) Death	
14.	_____ decides shape of the bacterium.		01
	(i) MreB	(ii) FtsZ	
	(iii) Microtubules	(iv) All of the above	
15.	Streptomycin is _____ antibiotics.		01
	(i) Narrow spectrum	(ii) Broad spectrum	
	(iii) Cell wall inhibiting	(iv) Both, (ii) and (iii)	
16.	_____ technique can always differentiate between live and dead cells during measurement of microbial growth.		01

	(i) Taking OD of culture	(ii) Viable count	
	(iii) Direct Microscopy	(iv)None of the above	
17.	Tincture contains ____		01
	(i) Phenol	(ii) Alcohol	
	(iii) Iodine	(iv) Surfactants	
18.	_____ can be used to measure microbial growth.		01
	(i) Cell count	(ii) Cell mass	
	(iii) Cell activity	(iv)All of the above	
19.	Find out the incorrect statement.		01
	(i) Light microscopy uses magnetic field.	(ii) Light microscopy can also be referred to as optical microscopy	
	(iii) Higher NA means better resolving power of any lens.	(iv)Methylene blue is a basic stain.	
20.	Full form of HEPA is ____		01
	(i) High Efficiency Particle Air	(ii) High Efficiency Particulate Air	
	(iii) High Efficiency Pressure Air	(iv)None of the above	

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Date: 7/5/2018 Day: Monday Time: 10:30 am to 1:00 pm Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q - II Answer the following questions as directed. (Any 10)		20
1	Define: Microbiology	02
2	Differentiate: Prokaryotes versus Eukaryotes (at least 4 differences)	02
3	Short note: Sand culture method to preserve fungal cultures	02
4	What is Pasteurization?	02
5	Highlight contribution of Robert Koch	02
6	Differentiate between stains and dyes	02
7	Explain: growth rate and generation time	02
8	Why do we use oil immersion lens to observe bacteria in the microscope?	02
9	Classify: Microorganisms based on oxygen requirements for growth	02
10	Explain growth pattern when glucose and lactose both are supplemented in media.	02
11	Define: Probiotics, Prebiotics and Synbiotics	02
SECTION - II		
Q-III Answer the following questions as directed. (Any 6)		30
1	Differentiate between Gram positive and Gram negative bacteria (at least 5 differences)	05
2	Discuss concepts and experiments involved to support or reject the theories of Biogenesis and Abiogenesis	05
3	Explain in detail: Microbial growth curve	05
4	How to measure microbial growth by cell count techniques?	05
5	Draw and explicate Whittaker's Five Kingdom classification	05
6	Classify microbes based on carbon, energy and reducing potentials	05
7	Write a note on various isolation strategies.	05